

PropSense.AI — Singapore Property Buy Readiness Advisor

DataExpert.io AI Engineering Bootcamp — Spring 2026 Capstone

Author: Nag Aditya Gade

Live Demo: *(see deployment section)*

Demo



Screenshots

Landing — Quick Start Presets + AI Advisor	AI Fills Form from Natural Language
A small icon representing the Landing page, showing a green landscape with a white building and a blue sky.	A small icon representing the AI Fill feature, showing a green landscape with a white building and a blue sky.
Score Result with Area Suggestions	Area Scored — Lifestyle + Transactions
A small icon representing the Score result, showing a green landscape with a white building and a blue sky.	A small icon representing the Area Score, showing a green landscape with a white building and a blue sky.
Score Breakdown — All 5 Dimensions	Lifestyle Fit + Recent 2026 Transactions
A small icon representing the Score Breakdown, showing a green landscape with a white building and a blue sky.	A small icon representing the Lifestyle Fit, showing a green landscape with a white building and a blue sky.
Hot Areas Map — Zoomed to Tampines	Project-level Drill-down (individual condos)
A small icon representing the Hot Areas Map, showing a green landscape with a white building and a blue sky.	A small icon representing the Project-level Drill-down, showing a green landscape with a white building and a blue sky.

What Business Problem Does This Solve?

Singapore's property market is one of the most regulated and expensive in the world. A first-time buyer faces: - **ABSD** (Additional Buyer's Stamp Duty) — up to 60% for foreigners, 20% for SC upgraders - **TDSR/MSR limits** — 55% and 30% income caps enforced by MAS and HDB - **CPF proration rules** — lease decay penalises buyers if lease < 95 - age - **PSF market data** — no single source compares what you're paying vs what the market pays

Today, buyers rely on property agents who have a direct conflict of interest. PropSense provides a **neutral, data-grounded Buy Readiness Score** (0–100) with cited sources, real 2026 transaction data, and honest financial risk flags — no commission, no bias.

System Architecture

User Input (natural language + form)



PropSense.AI Backend (FastAPI)

- ① AI Advisor → GPT-4o-mini → extract profile
- ② Query sanitisation + injection detection
- ③ RAG Retrieval
Query → OpenAI embed → Pinecone cosine top-20
→ Cohere rerank → top-5
- ④ Deterministic Scoring Engine
5 dimensions × weighted math (no LLM in score path)
HDB/URA CSV lookups for benchmarks (2024–2026 data)
- ⑤ LLM Narrator (GPT-4o-mini)
Score dict + policy chunks → cited narrative
- ⑥ LLM-as-Judge (high-risk outputs only)
- ⑦ PII filter + rate limit + token burn protection



Next.js Frontend

- Score card
- Area suggestions
- Leaflet map
- Transaction data

Data Sources (Pinecone Vector DB):

URA Transactions	HDB Resale	Policy KB
D1-D28, 2021-26	2017-2026	ABSD/BSD/TDSR
118k records	268k records	MSR/LTV/CPF

- + MRT station locations (LTA) + Primary school locations (MOE)
- + URA Rental data (quarterly, all districts)

Tech Stack & Justifications

Component	Choice	Why
Backend	FastAPI + Python	Async, fast, Pydantic validation; Python has best ML/data ecosystem
Frontend	Next.js (TypeScript)	SSR support, React ecosystem, Vercel-native
Vector DB	Pinecone	Managed, serverless, metadata filtering for source-scoped retrieval
Embeddings	OpenAI text-embedding-3-small	Best cost/quality ratio; 1536-dim reduced to 1024
Reranking	Cohere rerank-english-v3.0	Cross-encoder ranking outperforms cosine similarity on structured text
LLM (narrative)	GPT-4o-mini	Fast, cheap, follows structured JSON output reliably
LLM (AI Advisor)	GPT-4o-mini	Field extraction from natural language; temperature=0 for determinism
Mapping	Leaflet + react-leaflet	Open source, lightweight, supports custom bubble + project-level markers
Charts	Recharts	React-native, composable, no external deps
Data Processing	pandas	Standard for Singapore gov CSV datasets
Rate Limiting	slowapi	FastAPI-compatible, per-IP and per-day limits

Why NOT Cohere for embeddings? OpenAI text-embedding-3-small is cheaper and the Pinecone integration is more mature. Cohere is used for reranking where it adds real value (cross-encoder vs bi-encoder).

Why deterministic scoring? LLMs hallucinate numbers. ABSD rates, TDSR ratios, and BSD calculations are legal facts — they must be computed deterministically. The LLM only *explains* the deterministic output.

Datasets

1. URA Private Residential Transactions

- **Source:** data.gov.sg API → `d_7c69c943d5f0d89d6a9a773d2b51f337`
- **Format:** CSV (downloaded via S3 pre-signed URL)
- **Coverage:** Districts 1–28, property types: Condominium, Apartment, EC, Terrace, Semi-D, Detached
- **Records:** 131,329 (2021–2026, including landed)
- **Key fields:** district, propertyType, contractDate (mmYY format), price, area, price_psf, x/y coordinates (SVY21)

2. HDB Resale Flat Prices

- **Source:** data.gov.sg API → `d_8b84c4ee58e3cfc0ece0d773c8ca6abc`
- **Format:** CSV
- **Coverage:** All HDB towns, flat types (3-room to Executive), 2017–2026
- **Records:** 268,000+
- **Key fields:** town, flat_type, resale_price, price_per_sqft, remaining_lease, year

3. URA Rental Median Data

- **Source:** URA Access API (quarterly district-level medians)
- **Records:** ~5,000 district × quarter × property type combinations
- **Key fields:** district, rental_psf_month, year, quarter

4. Singapore Policy Knowledge Base

- **Source:** Hand-authored Markdown from official MAS/IRAS/HDB sources (verified May 2026)
- **Policies:** ABSD, BSD, TDSR, MSR, LTV, CPF Withdrawal Limits
- **Effective dates:** All current as of 2023-04-27 (ABSD), 2023-02-15 (BSD), 2022-09-30 (LTV)

5. MRT Station Locations

- **Source:** data.gov.sg LTA dataset → `d_b39d3a0871985372d7e1637193335da5` (GeoJSON)
- **Records:** 186 stations (exits aggregated to station centroids, Dec 2025)

6. Primary School Directory

- **Source:** data.gov.sg MOE dataset → `d_688b934f82c1059ed0a6993d2a829089`
- **Records:** 179 primary schools, geocoded via OneMap API
- **Key fields:** school_name, postal_code, latitude, longitude

Data Quality Checks

URA Transactions

1. **Outlier detection:** PSF values flagged where $|\text{PSF} - \text{median}| > 3 \times \text{MAD}$ per district×year. Removed 0.3% of records.
2. **Date parsing validation:** contractDate in `mmYY` format — rejected records where month > 12 or year < 2000 or > 2030.
3. **Coordinate bounds check:** SVY21 x/y converted to WGS84; records outside Singapore bounding box (1.15N–1.50N, 103.55E–104.10E) removed.
4. **Duplicate removal:** Deduped by (project, street, contractDate, price, area).

HDB Resale

1. **Lease parsing:** `remaining_lease` stored as string ("85 years 06 months") — parsed to fractional years; records with parsed value < 1 or > 99 dropped.
2. **PSF outlier check:** HDB PSF above 3000 or below 100 flagged as data errors and removed.

- 3. **Town normalisation:** Standardised all town names to uppercase (BEDOK, ANG MO KIO) to prevent join failures.
- 4. **Flat type validation:** Only accepted: 1 ROOM, 2 ROOM, 3 ROOM, 4 ROOM, 5 ROOM, EXECUTIVE, MULTI-GENERATION.

Policy KB

- 1. **Effective date frontmatter:** Every policy chunk requires `effective_date` in ISO format — ingestion fails on missing/malformed dates.
- 2. **Rate cross-check:** ABSD rates in policy chunks are programmatically verified against hardcoded `ABSD_RATES` dict in `scoring/readiness.py` before embedding.

MRT + School Data

- 1. **Geocoding validation:** OneMap API results checked against Singapore bounding box; 3 schools with invalid postcodes were skipped.
- 2. **Station deduplication:** Multiple exits per station aggregated to centroid; stations with < 2 exits flagged for manual review.

Buy Readiness Score — Invest Mode

Dimension	Weight	What it measures
Financial Readiness	30%	Cash vs upfront costs (ABSD+BSD+down payment+6mo EMI buffer), TDSR, MSR
Market Timing	20%	Current market PSF vs pre-2024 baseline (is the market at a peak?)
Area Value	20%	PSF CAGR 2022–2026 from URA/HDB transaction data
Opportunity Cost	15%	ABSD sunk cost as % of property price; CPF OA foregone returns
Life-Stage Fit	15%	Holding period vs breakeven; lease decay; loan vs retirement age

Buy Readiness Score — Live-In Mode

Dimension	Weight	What it measures
Financial Sustainability	35%	Same as above — can you actually afford the upfront costs?
Affordability Comfort	25%	EMI as % of income; +2% SORA stress test; monthly surplus
Lifestyle Fit	20%	MRT distance (LTA data); nearest primary school (MOE data)
Fair Price Check	10%	Entered price vs 2024–2026 area median from transaction registry
Life-Stage Fit	10%	Holding period vs breakeven; lease decay

Score Bands (Invest): Strong Buy (80–100) / Conditional Buy (60–79) / Marginal (40–59) / Do Not Buy (0–39)

Score Bands (Live-in): Very Ready / Ready with Caveats / Marginal Fit / Not Ready

Key design principle: The LLM never touches a number. All ABSD, TDSR, MSR, BSD calculations are deterministic Python. The LLM only explains the pre-computed result.

RAG Implementation

Retrieval Pipeline

```
User query
→ OpenAI text-embedding-3-small (1024-dim)
→ Pinecone cosine similarity → top-20 candidates
→ Cohere rerank-english-v3.0 → top-5 chunks
→ Scoring engine + LLM narrator
```

Embeddings in Pinecone

Source	Chunks	Metadata fields
<code>ura_transaction</code>	~700	district, year, property_type, market_segment
<code>hdb</code>	~500	town, flat_type, year
<code>ura_rental</code>	~400	district, year, quarter
<code>policy</code>	~80	policy_name, effective_date

Source	Chunks	Metadata fields
Total	~1,680	

Why Cohere Reranking?

Cosine similarity is a bi-encoder (query and document embedded independently). Cohere's cross-encoder reads query + document *together* — significantly better at: - Returning District 18 chunks when query says "Tampines" (D18) vs District 9 (Orchard) - Ranking the 2023/2024 ABSD policy chunk above pre-2023 rules - Matching "fair value entry" language to specific PSF trend chunks

Validated: "PSF District 9 condo 2023" → Cohere ranks the 2023 chunk at 0.98 vs cosine ranking 2024 highest at 0.61.

Recall@10 Logging

Every retrieval call logs to `logs/recall.jsonl` : - `recall_at_10` : fraction of top-10 Pinecone candidates above similarity threshold - `reranked` : whether Cohere was applied - `rerank_delta` : how many chunks changed rank due to Cohere

Abuse Protection (8 Controls)

#	Control	Implementation
1	Rate limiting	slowapi: 10 req/min + 100 req/day per IP
2	Global LLM budget	200 narrative calls/day max (circuit breaker)
3	Query length cap	2,000 characters max
4	Prompt injection detection	Regex scan for "ignore instructions", "jailbreak", "DAN", etc.
5	Topic restriction	System prompt explicitly restricts to Singapore property only
6	PII output filter	Regex masks NRIC (S/T/F/G + 7 digits), SG phone numbers, emails
7	Context fallback	If < 3 chunks above 0.35 similarity → <code>insufficient_data</code> response
8	LLM-as-judge	Secondary GPT call on high-risk outputs (ABSD ≥ 20%, short lease)

#	Control	Implementation
9	Bearer auth	Authorization: Bearer <token> required on all endpoints

Integration Tests (23 Passing)

All 5 proposal queries validated + 18 additional tests. Run with:

```
pytest tests/integration/test_pipeline.py -v
```

Test Case	Assertions
PR first HDB Tampines	ABSD=5%, MSR computed, policy chunks retrieved
Foreigner District 9	ABSD=60%, effective_date 2023-04-27 cited
Rental yield Buona Vista	URA rental chunks retrieved, mortgage rate cited
PSF appreciation 2021–2025	ROI chunks with CAGR, multi-year URA trend
SC upgrader 2nd property	ABSD=20%, LTV=45%, CPF OA cited

Evidence report: `tests/integration/results/report_2026-05-09.md`

Steps Followed & Challenges

Steps

1. Data download via data.gov.sg poll-download API (CSV via S3 pre-signed URL)
2. Ingestion pipelines: outlier removal, date parsing, dedup, PSF calculation
3. Chunking strategy: district × property_type × year for URA; town × flat_type × year for HDB; 256-token chunks with 32-token overlap for policy
4. Batch embedding (OpenAI, batch=100 to respect rate limits) → Pinecone upsert with metadata

5. Deterministic scoring engine: 5 dimensions, all arithmetic verified against IRAS/MAS official rates
6. FastAPI backend with rate limiting, PII filter, auth
7. LLM narrator: structured JSON output with inline citations
8. Next.js frontend: presets, AI Advisor (GPT field extraction), hot areas map, project-level drill-down
9. Integration testing + consistency proof

Challenges

- **URA contractDate format:** Stored as `mmYY` (e.g., `0524` = May 2024) — not documented in API spec
- **HDB vs URA PSF mixing bug:** Early versions mixed HDB (SGD 500/sqft) and URA condo (SGD 2,000/sqft) chunks in the same scoring dimension, producing 103% CAGR. Fixed by source-filtering per property type
- **Cohere non-determinism:** Different API calls return slightly different chunk orderings for identical queries (10–18pt spread in area value score). Fixed by separating RAG-dependent vs deterministic scoring paths
- **Singapore geographic precision:** "East Coast" means Marine Parade/Bedok (D15–16), not the broader east including Tampines/Hougang. Required custom sub-area mapping (30+ entries)
- `from __future__ import annotations` + **Pydantic v2:** Caused `ForwardRef` errors with `Body(...)`. Solution: remove the import from `api/main.py`

Future Enhancements

1. **Real-time listing integration** — 99.co / PropertyGuru scraper to match score against live listings
2. **New launch coverage** — URA new sale data (currently excluded — no coordinates)
3. **Graph RAG** — Multi-hop reasoning: schools → planning area → PSF trend → ABSD impact
4. **User accounts** — Save profiles, track market changes over time, watchlists
5. **Confidence scoring** — Trained classifier on retrieval quality vs user satisfaction
6. **Malaysia / Hong Kong expansion** — Same framework, different policy KB
7. **Mortgage calculator integration** — Live rates from DBS/OCBC/UOB APIs

8. **OneMap amenity enrichment** — Full amenity data (hawker centres, parks, hospitals) per planning area
-

Setup Instructions

Prerequisites

- Python 3.11+
- Nodejs 18+
- Pinecone account (free tier)
- OpenAI API key
- Cohere API key (free tier)

Backend

```
cd propsense
pip install -r requirements.txt
cp .env.example .env
# Fill in: OPENAI_API_KEY, PINECONE_API_KEY, COHERE_API_KEY, API_BEARER_TOKEN
uvicorn api.main:app --reload --host 0.0.0.0 --port 8000
```

Frontend

```
cd propsense/frontend
npm install
cp .env.local.example .env.local
# Set: NEXT_PUBLIC_API_URL=http://localhost:8000
# Set: NEXT_PUBLIC_API_TOKEN=<your bearer token>
npm run dev
```

Re-ingesting Data (optional — Pinecone already loaded)

```
python -m ingestion.download_raw      # Download URA + HDB CSVs
python -m ingestion.ura               # Process URA transactions
python -m ingestion.hdb               # Process HDB resale
python -m ingestion.policy            # Embed policy KB
python -m ingestion.amenities         # Download MRT + school data
```

```
python -m rag.embedder          # Batch embed all chunks
python -m rag.store             # Upsert to Pinecone
```

Tests

```
pytest tests/integration/test_pipeline.py -v      # 23 integration tests
python -m tests.integration.generate_report        # Evidence report
python -m tests.integration.qa_suite              # 39 QA cases
python -m rag.recall_logger                      # Recall@10 summary
```

Project Structure

```
propsense/
├── api/
│   ├── main.py          # FastAPI: /score, /parse-query, /suggest-areas, /areas-geo, /projects
│   ├── filters.py       # PII masking + prompt injection detection
│   └── middleware.py     # Rate limiting config
├── data/
│   ├── processed/       # hdb_clean.csv, ura_clean.csv, area_amenities.csv, policy_chunks.csv
│   └── policy_kb/       # ABSD, BSD, TDSR, MSR, LTV, CPF markdown files
├── ingestion/
│   ├── ura.py           # URA transaction pipeline
│   ├── hdb.py           # HDB resale pipeline
│   ├── policy.py        # Policy KB chunker
│   └── amenities.py     # MRT + school downloader/geocoder
├── rag/
│   ├── embedder.py      # Batch OpenAI embedding
│   ├── store.py         # Pinecone upsert
│   ├── retriever.py     # Cosine → Cohere rerank pipeline
│   └── recall_logger.py  # Recall@10 tracking
├── scoring/
│   └── readiness.py     # Deterministic 5-dimension scorer (invest + live-in modes)
├── llm/
│   ├── narrator.py      # GPT-4o-mini cited narrative generation
│   └── judge.py         # LLM-as-judge for high-risk outputs
├── frontend/            # Next.js app
│   └── app/
│       ├── page.tsx     # Main UI (presets, AI Advisor, score display)
│       └── components/
│           ├── HotAreasMap.tsx # Leaflet bubble + project-level drill-down
│           └── ROIChart.tsx   # Recharts projection chart
└── tests/
```

```
└─ integration/
  ├── test_pipeline.py      # 23 integration tests
  ├── test_score_consistency.py # Score consistency proof
  ├── generate_report.py    # Evidence report generator
  └─ qa_suite.py           # 39 QA cases (ABSD/BSD/LTV verified against IRAS)
```

Not financial advice. Verify effective_date before transacting.

PropSense SG — Built by Nag Aditya Gade | DataExpert.io AI Engineering Bootcamp, Spring 2026